Contrast-First: Determinacy as Commitment in the Event Density Ontology

Allen Proxmire

June 2026

Contents

Contrast-First: Determinacy as Commitment in the Event Density Ontology

Allen Proxmire — June 2026

Abstract

Event Density (ED) is usually presented through its machinery: thirteen primitives, two kernels, a coarse-graining theorem, a registry of derived structures. This paper steps beneath the machinery to ask what kind of thing ED *is*. The answer offered here is that ED is a **contrast-first** ontology. Its primitive is neither the *event* (a thing) nor the *relation* (a between), but **contrast** — a difference, a gradient, a participation pattern — from which both events and relations co-arise. The event-versus-relation dichotomy, which forces a choice between substance ontology and pure relationalism, is a false binary; ED dissolves it.

The paper makes one further, sharper claim. A relational ontology owes an account of how anything becomes *determinate* — how the relational flux yields particular facts. ED has a concrete answer that the relational-ontology tradition lacks: **commitment** (primitive P11), the substrate's irreversible event

in which one contrast is actualized as fact. Commitment does not create determinacy from nothing; it converts *relational potential* into *irreversible fact* — a substrate-level analogue of quantum collapse. From contrast-first plus commitment, two prohibitions follow that the framework must honor at every scale: **no determinate micro-fact in isolation**, and **no determinate macro-value (a physical constant) without global participation**. Together these read as a single principle — *determinacy is relational at every scale* — and they make ED's "value-inherited" methodological stance principled rather than apologetic: in a relational world, holistic quantities are not the kind of thing a substrate can produce in isolation.

The paper names its debts plainly. The arrow of determinacy enters as a *posited* irreversibility (P11) rather than a derived one: commitment actualizes facts *given* the arrow; it does not explain the arrow. And the perfect universality of quantities like electric charge is reconciled only by a hypothesis — that such quantities are discrete, topological invariants — offered here as an open prediction, not a result.

ED is situated against Whitehead's process metaphysics, Rovelli's relational quantum mechanics, and the ontic structural realism of Ladyman and French; its distinctive contribution is commitment as a concrete, discrete, already-load-bearing determinateness mechanism. No claim is made that this derives any physical constant. The aim is to state, clearly and defensibly, the ontology ED has been using all along.

---

1. Introduction

Foundational physics inherits a very old quarrel about what is basic: *things* or *relations*. Substance ontologies put things first — particles, fields, points — with relations as arrangements among preexisting relata. Relational ontologies invert this — relations or process are primary, and "things" are derivative knots in the relational fabric. The quarrel is alive in physics today, from relational interpretations of quantum mechanics to structural realism in the philosophy of science. Notably, *structure* is claimed by both camps — the substance view treats it as the arrangement of things, the relational view as the thing itself — which is the first hint that the dichotomy is not as clean as it looks.

Event Density sits on both sides of this line at once. Its substrate is built from *micro-events* (which sound like things) standing in *participation* relations (which sound like relations prior to things). Asked "which is more fundamental in ED, the event or the relation?", the framework's own vocabulary pulls both ways. This is usually treated as an imprecision to be resolved by picking a side.

This paper argues the opposite: the imprecision is a clue, and picking a side is the mistake. The event-versus-relation choice is a **false binary**. There is a third position — older than the binary, and a better fit for ED's actual machinery — in which event and relation **co-arise** from a more basic primitive. That primitive is **contrast**.

The intuition can be put in one line: *an event only events when given something to event with.* An event is not a self-standing object that then enters relations; it is the realization of a difference, and a difference requires two sides. A lone event is no event. This is the same intuition that, stated about quantities rather than events, says a dimensionless constant is meaningful only as a ratio — a relationship — and meaningless in isolation. "Two is more fundamental than one" is not a slogan here; it is the proposed metaphysics.

A relational ontology, however, incurs a famous debt the moment it is stated: if relations are basic and things are derivative, **how does anything become determinate?** How does the relational flux yield *this* fact rather than that one? Pure relationalism tends to leave this as an assertion — facts are simply "there." ED's contribution, the paper will argue, is to *not* leave it as an assertion. ED has a concrete operation — **commitment** — that converts relational potential into irreversible fact. Contrast supplies the ontology; commitment supplies the determinateness.

The paper proceeds as follows. §2 presents contrast as the primitive and confronts the obvious risk that "contrast" becomes a word that explains everything and therefore nothing. §3 develops co-constitution: how event and relation arise together. §4 presents commitment as the determinateness mechanism, in its refined form (actualization, not creation), and names where the arrow of time enters. §5 states the two prohibitions that contrast-first must enforce. §6 draws the payoff for ED's methodology: value-inherited as a principled epistemology. §7 is the honest frontier — exact quantization, the topological-charge hypothesis, and the boldest exposure. §8 situates ED among existing frameworks. §9 concludes with an explicit statement of what is and is not claimed.

---

2. Contrast as the Primitive

By **contrast** we mean the minimal unit of difference: a gradient between densities, a sign-flip across a boundary, a participation pattern that distinguishes one channel from another, a change of state from one moment to the next. Contrast is concrete in ED — it is not an abstract logical "distinction" but a substrate-level quantity (a difference in event density, an edge in the participation graph, a polarity transported between loci, a state-change across a commitment). Wherever ED has structure, that structure is a configuration of contrasts.

The claim *contrast is the primitive* must immediately answer a hard objection, because if it cannot, it is empty. The objection: one can append "is a contrast" to every concept in the framework — a relation is a contrast between densities, an event is a realized contrast, a boundary is a persistent contrast, a constant is a contrast ratio — and a word that describes *everything* describes *nothing*. A primitive earns its place only if it **restricts** — if it tells you what cannot happen. A universal solvent is not a foundation.

We accept this test and meet it in §5, where contrast-first is shown to forbid two specific things. Here we only fix the status of the primitive: contrast is offered not because everything can be *called* a contrast, but because contrast is what is left when one refuses to grant primacy to either side of the event/relation binary. If neither the relatum nor the relation is prior, what remains as basic is the *difference itself* — the having-of-two-sides. That is contrast. Its content as a primitive is exactly the prohibitions it underwrites, not the breadth of things it can be used to redescribe.

This locates ED in a recognizable lineage without collapsing into it. Whitehead, in *Process and Reality*, uses "contrast" as a technical category — the mode in which diverse elements are synthesized into a single occasion. Spencer-Brown's *Laws of Form* takes the drawing of a distinction as the primitive act; Bateson's "a difference that makes a difference" makes difference the unit of information. ED's contrast is a physical, substrate-level cousin of these: difference taken as fundamental, with the framework's specific machinery (gradients, participation, the stability functional, commitment) as what makes the difference *do* anything.

---

3. Co-Constitution: Event and Relation Arise Together

If contrast is primitive, then events and relations are not two competing candidates for "the basic thing" but two **aspects** of one process, distinguished by phase.

In ED's machinery a chain — a persistent micro-event sequence — does not first exist as a determinate object and then enter participation. Before commitment, a chain *is* a participation pattern: its identity is spread across channels, multiple continuations are live, no single outcome is fixed. This is the **relational phase**. It is genuinely structured — the pattern is a specific pattern — but it is not yet a determinate fact. After commitment, one continuation is fixed as *the* realized event: the **event phase**.

So the order of being is not "event, then relation" nor "relation, then event." It is:

contrast $\rightarrow$ relational phase (participation) $\rightarrow$ commitment $\rightarrow$ event phase (fact)

Event and relation are two phases of a single becoming — the relational phase before commitment, the determinate phase after. But this temporal order is not an order of priority. That participation precedes the committed event *in time* makes the relation no more fundamental than a question is more fundamental than its answer; neither is intelligible without the other. A relation with no terms to relate is empty; a term with no relations is, in ED, literally undefined — there is no determinate event in isolation (§5). This is co-constitution: the terms and the relating arise together, against contrast as their shared ground.

This dissolves the dichotomy rather than choosing within it. ED is not a substance ontology with relational vocabulary, and not a pure relationalism that dispenses with relata. It is a process ontology in which the *thing* and the *between* are co-emergent, with a definite mechanism marking the transition between phases. That mechanism is the subject of §4.

---

4. Commitment as Determinateness

The relational phase is, by construction, indeterminate-as-fact: participation spread across channels, no single outcome fixed. Something must convert this into the determinate facts that physics describes. In ED that something is **commitment** (primitive P11): the irreversible substrate event in which a chain's multi-channel participation collapses to a single channel, with phase-randomization, and is never un-collapsed.

It is tempting to state the role of commitment too strongly — to say commitment *creates determinacy out of pure indeterminacy*. That overstatement does not survive scrutiny, and it is worth seeing why, because the corrected version is both more honest and more powerful.

The objection to the strong claim: for a chain to commit, there must already *be* a chain — a specific participation pattern — to undergo commitment. So something is there before commitment; determinacy is not conjured from nothing. The pre-commitment pattern is a real structure.

The correction distinguishes **two kinds of determinacy**:

- **Relational (potential) determinacy** — the participation pattern itself. This *pre-exists* commitment. The pattern is a definite pattern; the relational phase is structured, not formless.
- **Factual (actual) determinacy** — *which* continuation becomes the realized, irreversible fact. This is what commitment produces.

Commitment therefore does not create determinacy *ex nihilo*; it **actualizes** a relational potential into an irreversible fact. Quantum mechanics already supplies the right picture — taken as an analogy of *structure*, not as a claim that the substrate is itself quantum (ED aims to *derive* quantum mechanics from the substrate, so the substrate is pre-quantum). A superposition is a perfectly *definite state* — a precise mathematical object — whose measurement *outcome* is nonetheless open until measurement fixes it. The state is settled; the outcome is not. The parallel runs:

> superposition (definite state, open outcome) $\to$ measurement $\to$ result (fixed outcome)
>
> participation pattern (definite pattern, open outcome) $\to$ commitment $\to$ event (fixed fact)

In both rows, what pre-exists is a definite relational structure with an unfixed outcome; the operation fixes the outcome, irreversibly. **Commitment is ED's substrate-level collapse: the moment a contrast becomes a fact.** Contrast is the primitive; commitment is the actualization.

Two honest points must accompany this.

The arrow enters as a posit. Commitment "makes" factual determinacy *because it is irreversible* — and irreversibility is posited (P11), not derived. The directionality of determinacy is therefore an *input*, a named primitive, not an output of the mechanism. This is not a defect; every theory posits something. But it fixes the scope of the claim precisely: *commitment actualizes determinacy given a posited irreversibility; it does not derive the arrow of time.* (Within the broader ED corpus, whether P11's irreversibility is fundamental or reducible to other arrow-supplying primitives is itself flagged as open. Nothing here turns on resolving that; the present claim needs only that the arrow is named, not hidden.)

Individuation is by position, not by intrinsic tag. If events carry no intrinsic properties (§7), what makes two events distinct? The answer is the one causal-set theory already uses: an event is individuated by its **position in the irreversible participation history** — which patterns fed into it, which channel it committed from, which chains it enables downstream. Commitment fixes that position irreversibly; that is what makes it *this* event. Identity is place in the record, not an intrinsic label. This works generically; the honest edge is perfect symmetry, where structurally identical events would fail to be distinguished by relational position alone — but a substrate that is a specific accumulated history of commitments generically breaks such symmetries, so the failure is non-generic. The edge is named, not hidden.

What makes commitment ED's own contribution, rather than borrowed philosophy, is that it is not introduced here for ontological effect. P11 is an existing, load-bearing primitive: it does the work in ED's derivations of the Born rule (via commitment phase-randomization), of measurement structure, of the collapse-like behavior the quantum arc requires. This paper does not invent a mechanism; it articulates the *ontological role* of a mechanism ED already uses to derive results. That is the difference between a determinateness *assumption* (which the relational tradition has) and a determinateness *mechanism* (which ED has).

---

5. The Two Teeth: What Contrast-First Forbids

A primitive earns its place by forbidding. We identify two things contrast-first forbids — and the smallness of that number is itself the point. Candidate prohibitions that turned out to be restatements of the thesis, or to belong to ED's *locality* structure rather than to contrast-first, were set aside; what remains are two genuine, independent restrictions. We do not claim these exhaust what contrast-first forbids — only that these two are real and survive scrutiny.

Tooth 1 — No determinate micro-fact in isolation. There is no determinate event, value, or property belonging to a single participant considered alone. Determinacy is conferred only at commitment, and commitment is inherently relational (a collapse *among* channels, fixing a position *in* a web). A lone event is no event. This is not a methodological caution; it follows from co-constitution plus commitment, and it is already encoded in ED's machinery.

Tooth 2 — No determinate macro-value without global participation. A physical constant's value is not a property a single participant or a local rule can carry. It is the equilibrium of the whole relational web. *Constants are contrast equilibria of the whole web.* One cannot read a global value off a substrate fragment any more than one can read a gas's temperature off a single molecule's trajectory: temperature is collective, and so — on this account — are the constants.

These are two ends of one principle:

> **Determinacy is relational at every scale.** Neither the smallest thing (a single event) nor the largest (a fundamental constant) is determinate in isolation. The micro end needs a contrast partner; the macro end needs the whole web.

"Two is more fundamental than one" thus holds at both ends: a lone event is no event, and a lone value is no value. The two teeth are the content of contrast-first — the specific things it rules out — and they are what answer the universal-solvent objection of §2. Contrast-first is not the claim that everything can be called a contrast; it is the claim that *nothing is determinate alone*, at any scale.

---

6. The Payoff: Value-Inherited as Principled

ED's stated methodology splits every result into what is **form-FORCED** (derived from the primitives and declared postulates) and what is **value-INHERITED** (the numerical value, taken from empirical matching rather than derived from first principles). It would be easy to read "value-inherited" as a confession of incompleteness: a fuller theory, one might think, would derive the numbers too.

The two teeth turn this reading on its head. If a constant's value is a property of the *whole relational web* (Tooth 2), then no substrate-level or local description can derive it — **in principle**, not merely in practice. The value is a holistic quantity; the substrate is a local description; asking the substrate to output the number is a category error of the temperature-of-one-molecule kind.

So value-inherited is not ED falling short of a derivation it ought to have. It is ED *correctly drawing the line* between what is local-and-structural (the *form* of a law — derivable, hence form-FORCED) and what is global-and-relational (the *value* — holistic, hence value-INHERITED). **Value-inherited is not a weakness; it is the epistemology of a relational world.** A theory that claimed to grind every number out of a substrate would, in this view, be making a category error — and, not incidentally, would be indistinguishable in form from numerological "derivations" of constants, which insert the target value and recover it dressed in new notation.

This reframe carries its own discipline, and the paper states it to avoid the obvious abuse. "It is relational, therefore inherited" is licensed *only* where a quantity is genuinely holistic. It must not become a reflex that forecloses every derivation. Where a quantity is local and structural, ED should still attempt to derive it; the *form* of a law remains fair game, and indeed is what ED claims to force. The relational reframe explains why certain *values* are inherited in principle; it is not a blanket exemption from doing physics. The honest division of labor is: structure is derivable, holistic values are inherited, and which side a given quantity falls on must be argued case by case.

---

7. The Honest Frontier

The relational account faces its hardest test where quantities are *perfectly universal*. Electric charge does not vary: every electron carries the same charge, in every context and at every accessible energy, to within about one part in 10^{21}. Spin is exactly one-half. This is the sharp problem for contrast-first, because relational quantities are supposed to *vary* with context — a coupling runs with energy, a temperature depends on the gas. A quantity that never varies looks exactly like an intrinsic, absolute property — the very thing contrast-first denies. If charge is relational, why is it everywhere identical?

The resolution requires splitting "relational property" into two flavors:

- **Discrete relational invariants.** A topological charge — a winding number — is relational (a global feature of a configuration relative to a reference) yet can take only integer values, and so is *exactly* universal. Its exactness comes from **discreteness**, not from intrinsic substance. Charge quantization from topology is not exotic: it is the content of the Dirac monopole argument and of the topological charges carried by solitons. Spin is the same story from representation theory — spin is *how a state transforms under rotation* (a relation), quantized by the structure of the rotation group. Spin is, if anything, the easiest case for relationalism: even standard physics agrees it is a transformation property, not a substance.
- **Continuous relational quantities.** Masses and couplings, which run with scale and are set by coupling to the vacuum — relational in the *web-equilibrium* sense of Tooth 2, hence running and inherited.

The payoff of the split is that it maps onto the two teeth: discrete invariants are the *micro/exact* end, continuous quantities the *macro/inherited* end. The *exactness of charge* and the *inheritedness of the couplings* become two faces of one relational ontology — and, suggestively, the exactness of the discrete invariants would trace to the **discreteness of the ED substrate** (primitive event-discreteness), so that quantization is a consequence of substrate structure rather than an imposed axiom.

This is where the paper must be most careful, because this is its boldest and least-paid frontier. Two debts are named explicitly:

1. **A debt of derivation (the topological-charge hypothesis).** It is one thing to say the relational ontology *predicts* charge to be a discrete/topological invariant; it is another to *show* that the ED participation structure actually carries a winding-number-like invariant equal to electric charge with the correct quantization. ED has not shown this. The "gradient defect" picture of charge is suggestive, not derived. The honest status is therefore a **prediction**, not a result: *contrast-first requires exactly-universal quantities to be discrete relational invariants; whether the substrate realizes charge this way is open, and is the load-bearing test of the proposal.* This prediction is falsifiable in a useful direction — if charge cannot be given a topological/relational reading, contrast-first is in trouble, because charge's perfect universality would then stand as a genuinely intrinsic absolute property.

2. **A debt of metaphysical commitment (thin relata).** Denying intrinsic properties commits ED to events with no intrinsic individuating quality — distinguished purely by structural position

in the participation history (§4). This is a coherent and named position (the "thin relata" of ontic structural realism), and ED's commitment mechanism supplies the irreversible "thisness" that bare structuralism lacks. But it is the boldest metaphysical claim in the paper, and it must be made out loud rather than assumed.

Naming these debts is not a weakness of the proposal; it is the proposal taking its own discipline seriously. The frontier is where contrast-first could either unify (the exactness/inheritance split) or fail (if charge resists a relational reading). Both outcomes are informative.

---

8. Relation to Existing Frameworks

Contrast-first is not novel in being relational; it is novel in *how* it secures determinacy. It is worth saying precisely what ED inherits and what it adds.

Whitehead (process metaphysics). ED's substrate is strikingly close to Whitehead's: micro-events map to "actual occasions" of becoming, and participation maps to "prehensions" — an occasion is constituted by its graspings-of other occasions, not a substance that subsequently relates. Whitehead even uses "contrast" as a technical category. What ED adds is a *sharp* mechanism of becoming-determinate. Whitehead's "concrescence" is notoriously hard to pin down; ED's commitment is discrete, irreversible, and already does derivational work.

Rovelli (relational quantum mechanics). Relational QM holds that a system's state is defined only relative to another system — there are no absolute states. ED agrees at the level of the relational phase: participation is relative, and there is no observer-independent pre-commitment fact. What ED adds is the *transition*. Relational QM leaves every fact relative to some system, with no point at which a fact becomes simply settled. ED's commitment is that point — the irreversible substrate event at which a relational potential becomes a fixed fact of the shared history, the same for all.

Ladyman and French (ontic structural realism). OSR holds that structure is fundamental and "things" are secondary or derivative. Contrast-first is a version of this, with the two teeth as its concrete content. What ED adds, again, is the determinateness mechanism: OSR holds that there are facts within structure but offers no operation by which a particular fact becomes fixed. ED's commitment is exactly such an operation — and it carries the "thisness" (individuation by irreversible position) that thin-relata structuralism is often accused of lacking.

The common thread is that the relational tradition is rich with *what is fundamental* (relations, structure, process) and short on *how determinacy arises*. That is the gap contrast-first fills. The distinctive ED claim is therefore narrow and defensible: **a relational ontology with a concrete, discrete, irreversible, already-load-bearing determinateness mechanism — commitment — as the operation that turns relational potential into irreversible fact.** ED's broader intellectual neighborhood — 't Hooft's cellular automata, Wolfram's hypergraph rewriting, the causal-set program — shares the discrete-substrate impulse; ED's specific addition to *that* neighborhood is the same: the commitment event as the hinge of determinateness, and the contrast-first reading of what the substrate's structure *is*.

---

9. Conclusion

This paper has argued that Event Density is, at bottom, a **contrast-first** ontology. Contrast — difference, gradient, participation pattern — is the primitive; events and relations co-arise from it as the *after* and *before* of a single becoming. The determinateness that physics requires is supplied by **commitment** (P11), which does not create facts from nothing but *actualizes* a relational potential into an irreversible fact — a substrate-level collapse. From this follow two prohibitions, the two teeth: no determinate micro-fact in isolation, and no determinate macro-value without global participation. Together they say *determinacy is relational at every scale*, and they make ED's value-inherited methodology a principled epistemology rather than an apology: holistic values are not the kind of thing a substrate produces in isolation.

The proposal's debts are named rather than hidden. The arrow of determinacy enters as a *posited* irreversibility (P11), not a derived one: commitment actualizes determinacy *given* the arrow; it does not explain the arrow. The exact universality of quantities like charge is reconciled only by the hypothesis that such quantities are discrete relational (topological) invariants — a *prediction* of contrast-first, and an open derivation ED has not delivered. And the denial of intrinsic properties commits ED to thin relata individuated by irreversible position — coherent, but the boldest claim made here.

What is **not** claimed should be equally clear. Nothing in this paper derives a physical constant; on the contrary, the account explains *why* certain values cannot be derived from the substrate alone. Nothing here is a result in physics; it is a statement of ontology — the kind of thing that clarifies what a framework is and what its honest commitments are. And the relational reading is not a license to stop doing physics: the *form* of a law remains derivable and is what ED claims to force; only genuinely holistic *values* are inherited in principle.

The contribution, stated plainly: ED supplies the missing operation — commitment — and reads its substrate's structure as configurations of contrast. *An event only events when given something to event with* — and it becomes a fact when, and only when, a contrast commits.

Open Questions (Register)

1. **Topological charge (debt of derivation).** Does the ED participation structure carry a discrete/topological invariant equal to electric charge with correct quantization? Contrast-first *predicts* yes; ED has not shown it. Load-bearing test of §7.
2. **The arrow (scope of P11).** Commitment's determinateness rests on posited irreversibility. Whether P11's arrow is fundamental or reducible (to time-homogeneity plus kernel retardation) is open; the present account requires only that it be named, not derived.
3. **Thin-relata individuation at perfect symmetry.** Relational individuation by participation-history position is generic but fails for exactly symmetric configurations; the claim that accumulated commitment history generically breaks such symmetries should be made precise.
4. **Minimality of "contrast."** Whether "contrast" is genuinely a single primitive or a working consolidation of distinct substrate notions (gradient, participation, polarity) parallels the open minimality questions about the thirteen primitives.